

ADR 0105 — The conflict model is a crate, and the terminal only draws it

Status: Accepted — implemented, mutation-proved two ways per invariant (14/14), full gate green **Date:** 2026-09-02 **Issues:** [#462](#) — M10.07, conflict resolution in the terminal, on the shared conflict model **Follows:** ADR 0063 (the conflict read model), ADR 0064 (whole-side resolution), ADR 0066 (why the client-side conflict logic is framework-free), ADR 0068 (a conflict's shape is read from `kind`), ADR 0069 (the `conflict-v1` token), ADR 0101 (the session boundary is a crate), ADR 0103 (the selection belongs to the session) **Supersedes:** nothing · **Superseded by:** nothing

Context

#462's goal is one sentence, and every hard thing about the issue is in it:

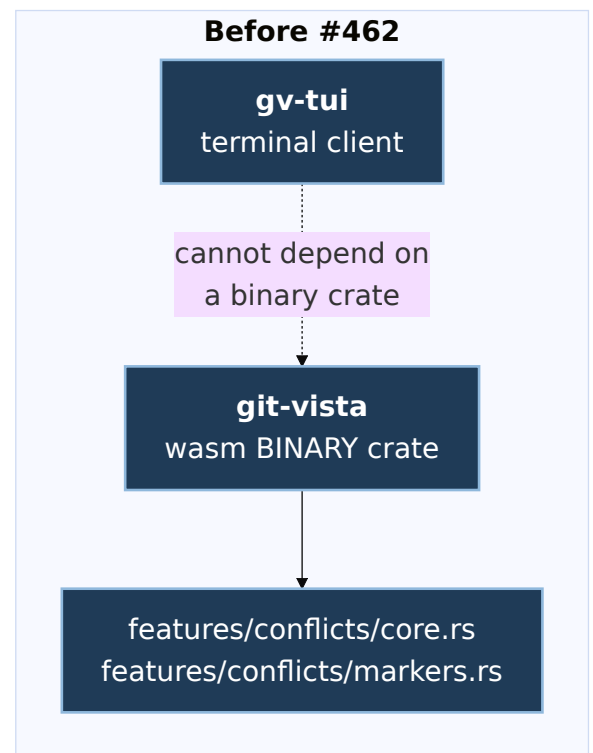
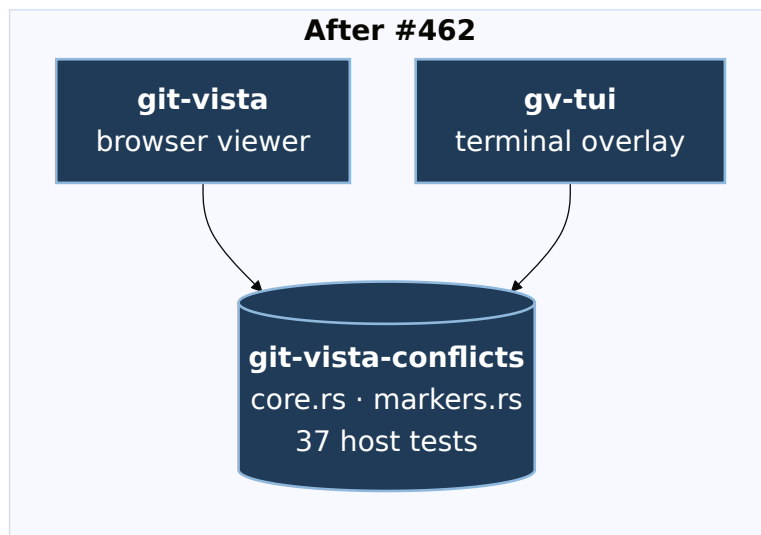
Inspect and resolve conflicts from the TUI, **using the same model M4.31 shipped — not a second conflict implementation.**

The model was already there and already good.

`features/conflicts/core.rs` held the four-pane mapping, the six pane states and the `ResolutionSurface` that decides which whole-file controls may be offered; `markers.rs` held the marker-file parser and composer. Both were framework-free and host-tested, on purpose, for the reason ADR 0066 records: two of #428's four criteria are facts about **rendering**, and a mapping behind `# [cfg(target_arch = "wasm32")]` is a mapping `cargo test` never compiles.

One thing stood in the way, and it was structural rather than conceptual. Those files lived inside `git-vista`, which is a **binary**

crate built to wasm by Trunk. `gv-tui` had nothing to depend on. The choice was not "reuse or rewrite" — it was "move the model, or write it twice."



Decision

1. The model moves to `git-vista-conflicts`, a crate downstream of both core and protocol

`core.rs` reads `git_vista_core::diff` (the blob and worktree payloads) and `git_vista_protocol::conflict` (the stages). Folding it into either of those crates would break a dependency invariant each of them writes down: `git-vista-protocol`'s lib doc says it depends on neither, and that "`git-vista-core` does *not* depend on it, keeping the domain model free of transport concerns." A crate downstream of both breaks neither claim.

The precedent is one milestone old and exactly this shape: ADR 0101 extracted `git-vista-session` when `gv-tui` needed the session logic `git-vista-mcp` already had. **Alternative considered and rejected:** add `git-vista-protocol` to `git-vista-core`'s dependencies and put the module there. It is one line of manifest, and it silently reverses a documented architectural claim in two crates — the cheapest edit and the most expensive decision.

The move was a `git mv` plus a manifest. No logic was rewritten and all 37 of M4.31's host tests came along and still pass. **That is ADR 0066's dividend being collected**, and it is worth naming: had these decisions lived in the Leptos viewer, #462 would have had to

reimplement them, and the reimplementation would have been tested by nothing.

The `features/conflicts` module is deleted rather than kept as a re-export. With no code of its own left, a forwarding module only hides where the model now lives.

2. Exactly two widenings of the shared surface, each with a reason

The issue permits `markers.rs` to "move or gain a public surface... deliberately". Two did, and no more:

- `markers::Choice::describe()` — the four-way wording ("keeping both" and its siblings) was spelled out inline in `viewer.rs`. The terminal needs the same four words. Two lists of four strings in two crates drift, in a UI whose entire job is telling somebody exactly which version of a file they are about to keep.
- `core::ConflictPanels::pane_mut()` — the terminal folds blob answers back one at a time, tagged with the pane they were asked for. Without it, it would have written its own `match pane { Base => &mut self.base, ... }`, a second copy of the one mapping that says which field is which side. #612 found precisely that shape a week ago: `CherryPick` and `RevertCommit` both carry one `Commit0id`, so mapping one to the other compiles and renders the exact inverse operation.

3. The terminal draws what the model returned, and decides nothing

Every judgement in the overlay is delegated, and this is the criterion the issue actually cares about:

question	who answers
what state is this pane in	<code>PaneState::for_stage / with_content / result_pane_state</code>
what does that state say to a user	<code>PaneState::describe</code>
may <code>Take</code> ours be offered, and if not why	<code>ResolutionSurface::take_ours , Withheld::describe</code>
what shape of conflict is this	<code>ResolutionSurface::note</code>
may a line-level resolver open at all	<code>ResolutionSurface::text_resolution_allowed</code>
what blocks does the marker file hold	<code>markers::parse</code>
what content does a set of choices produce	<code>markers::compose</code>

`text_resolution_allowed` is read, never re-derived. It traces to `ConflictedFile::text_resolvable`, the identical call the server makes before executing a content resolution. #430 shipped a wrong sentence because that rule had two implementations; it still has one.

Proving that is harder than it looks, and the proof is worth recording. The predicate is three clauses — no typed reason, **and** both live sides actually text — and on every ordinary fixture the obvious local re-derivation (`not_text_resolvable.is_none()`) agrees with it. A test suite built only from ordinary fixtures would pass on the wrong copy. So the suite carries the documented **disagreement** case: a conflict whose wire payload has no `not_text_resolvable` but whose stage is flagged binary, where the protocol's own doc says the per-side flag wins. The naive re-derivation says yes there; the real predicate says no. That fixture is the whole difference between testing the rule and testing a fixture.

4. The four panes are a summary strip plus one full-width body, not a 2×2 grid

The frame's floor is 40 columns. Quartered, that is 20 columns of source per pane — every line of every version truncated, and the user reading the *shape* of four boxes rather than their contents. That is the same failure as a pane that draws an empty box: it looks like inspection and is not.

So all four panes are always **stated** — one row each carrying that pane's own `describe()` sentence, so "Not present on this side" is on screen whichever pane is being read — and the focused one is shown full width. `Tab` and `1 - 4` move between them. Every pane stays reachable, which is what #428's first criterion asks, and none is ever a blank box.

The same distinction is applied one layer in: a `Block::Conflict` whose `base` is `None` renders "no recorded ancestor in this marker file", never an empty ancestor section. Git omits the ancestor under the default merge style, and a blank section would claim a common ancestor existed and was blank — ADR 0063's rule, inside the editor.

5. A conflict write is paired with its own `POST /api/select`, every time

This is the decision least visible in the diff and most likely to matter.

Reads in this client address a repository explicitly with `?repo=`. `/api/resolve-conflict` addresses none: it goes through the planner, which acts on **this session's selection** (per-session since #588, ADR 0103; before that, per-process). A client that listed one repository's conflicts and posted a resolution without selecting would write to whichever repository the server launched with — and if that repository happened to have a conflict at the same path, the write would **succeed, in the wrong repository, silently**.

Selecting once when the overlay opens would fix the common case and leave a remembered fact that can go stale. Pairing the select with the write makes "the resolution lands where the user was looking" true by construction. `Active`, because `reject_if_read_only`

refuses `Visualize` and it is the only mode a write is legal in; and it is this terminal session's own selection, so the browser's is untouched.

The deeper fix is a protocol change and is deliberately not made here. The endpoint arguably ought to carry the repository, the way every read does. That is a wire-contract decision with its own blast radius, it belongs in its own issue, and #462 is not the place to make it. Recorded so the next reader knows the pairing is a mitigation with a known better answer, not the end of the thought.

6. The overlay owns the keyboard, because its editor takes every printable key

`keys::dispatch_conflict` is a second table rather than more arms in the first. It has to be: under one shared keymap, typing `q` into a file you were resolving would quit the program and throw the edit away, and `j` would scroll instead of appearing in the line. `Ctrl-C` is the single binding that survives insert mode — a terminal program you cannot interrupt is one somebody has to kill from another window.

`x` opens the overlay. `c` was the obvious mnemonic and is left alone: the working-tree slice (#459) is the natural owner of a commit key.

7. A hand-edit is authoritative from the first keystroke, and only from then

`Editor::hand_edited` is a separate flag from `buffer.is_some()`, and the distinction is the whole of #462's fourth criterion. Merely *opening* the composed text must not freeze the block choices — that would be the rule firing on somebody who did nothing. A typed character must. So:

- before any edit, re-entering the buffer **re-seeds** from the current composition, because until then the buffer is only a view of it and a stale seed would show a choice the user has since changed;
- after one, the buffer is the user's text: block choices go inert and say so, the buffer is never re-seeded, and the submission

carries the typed text rather than the composition.

Re-composing over somebody's typing is the worst failure available in an editor, because it is silent. The seed itself is never empty: it falls back to the marker file git actually wrote, since an empty buffer would offer "delete everything in this file" as the starting point of a resolution.

8. `authed_fetch_response`, because a 404 is information

`GET /api/worktree-file/{path}` answers 404 when there is no file at the path, and in a delete/modify conflict that is exactly what git left behind. Read through the ordinary JSON path, that fact arrives as "content could not be loaded" — a fault reported where nothing went wrong, which is the collapse `Stage::Absent` versus `Stage::Unreadable` exists to prevent, arriving one layer down in the transport. So `git-vista-session::retry` gains a sibling that hands back the status. It also returns whether the answer arrived on a freshly minted session, so `authed_fetch` keeps its exact "even after re-authenticating" wording rather than losing that distinction to the refactor.

9. The idempotency key is unique per press, not derived from the request

A key names **one user action**, and the server replays the recorded outcome for a key it has already seen. A key derived from the resolution's own content would make a second, deliberate attempt look like a retry of the first and replay its answer instead of running — for a refusal, that means being told again about a repository state that has since changed. The wall-clock nanosecond is in the key because the operation registry is durable across server restarts (#62): a bare counter restarts at 1 in a fresh `gv-tui` and collides with the previous run's keys, and a collision here is a write that silently does not happen.

Consequences

- One conflict implementation, two clients. A fix to the model reaches the browser and the terminal at once, and neither can drift.
- `git-vista-core` and `git-vista-protocol` keep the dependency invariants they document. The workspace gains a crate; that is the price.
- The write path depends on a select landing first. If `/api/select` is ever removed or changed, `gv-tui`'s conflict writes are affected — the pairing is named on `select_for_write` so the connection is findable.
- `Client::with` grew a third parameter. Every existing test passes a closure that panics if anything posts, which is louder than a benign stub.
- **`row_count` is a second implementation of the row layout**, arithmetic beside the visitor's walk, and it exists because a 2 MB conflicted file must not allocate thirty thousand rows per redraw. The honest way to hold two implementations of one thing is to pin them against each other, so `row_count_agrees_with_the_rows_actually_emitted_on_every_screen` fails the moment they diverge, on all three screens.
- Not attempted, and not pretended: rebase and cherry-pick sequence control beyond what conflict continuation needs (#461's territory), and the protocol change in decision 5.

Verification

Full gate green on the exact head. `cargo test -p gv-tui --bins`: **115 passed**, counted by name rather than read off the word "ok" — the `--lib` false-green this repository has lost cycles to twice does not apply here, but the counting habit does.

Fourteen mutations through `failure-atlas`, seven invariants at two ways each that fail differently — one removing the mechanism, one weakening it — and **all fourteen caught**:

invariant	removed	weakened
a hand-edit makes choices inert	guard deleted from <code>choose</code>	<code>apply_content</code> prefers the composition
the editor reads the server's flag	gate deleted	flag replaced by <code>note.is_some()</code>
an absent pane says so	summary strip narrowed to the focused pane	the sentence emptied to <code>" "</code>
<code>row_count</code> matches the walk	the note term dropped	the ancestor row dropped
a write selects its repository first	select deleted	select moved after the post
a 404 is an absent file	the 404 arm deleted	<code>NoFile</code> mapped to <code>Failed</code>
the stage triple and token are echoed	token re-minted locally	stages replaced with <code>[None; 3]</code>

One mutation is worth singling out. Replacing the eligibility flag with `note.is_some()` — a re-derivation that looks entirely reasonable — is caught **only** by the disagreement fixture in decision 3. Every other conflict test in the module stays green under it. That is the difference a single fixture made, and it is why "the tests pass" was not accepted as evidence for criterion 3 until this run.

Signed: max · 2026-09-02T19:10:00-04:00